L0 space

L1 length

L2 length of rod

L3 1.234 m ± 0.001

ISO 80000 & IEC 80000

QUANTITIES · UNITS · MEASUREMENT

ISQ in SMART

The International System of Quantities as an information service

Ronald Tse — Ribose

November 2026

A SERVICE CONTRIBUTED TO ISO/TC 12

1.234 m ± 0.001

History of the dataset

2021 — **BIPM proposes a “SMART” dataset to ISO**
BIPM and Ribose co-develop the ISO | IEC 80000 SMART dataset as a proposal project

2021–24 — **Manual encoding of all parts**
Every symbol, quantity and unit of Parts 2–13 encoded into one machine-readable dataset

2024–25 — **The dataset becomes the source of truth**
ISO 80000-2 and ISO 80000-4 revisions are driven from the ISQ dataset in ISO/TC 12

2026 — **From dataset to information service — and to ontology**
The ISQ Browser publishes the series as linked data; a measurement layer models quantities in the real world

DATASET

Two parts of the dataset: math and quantities

Mathematical notation

ISO 80000-2 — signs, symbols and math concepts, each with its machine-readable expression.

`m2-1.1` ± plus-minus

Quantities and units

Parts 3–13 — every quantity with ID, name, symbol, definition, unit and notes.

`t3-1.1` length l

- One ID space, one structure — **ISO 10241-1 terminology schema** under the whole dataset
- Cross-references between parts resolve as data, not prose

REVISIONS

The ISQ dataset is now used to revise the 80000 series

ISO 80000-2 and ISO 80000-4 are being revised now, using the ISQ dataset as the source of truth in ISO/TC 12. Future revisions are expected to continue from the dataset.

13
parts encoded
Parts 2-13

765
quantities
in the browser today

230
math entries
ISO 80000-2

82
units
sourced from UnitsDB

NATURE OF CONTENT

Two kinds of content in the 80000 series

Informative — Terms

Published on the OBP. Definitions and term entries, publicly browsable.

Normative — Quantities + Units

The “ID / Quantity / Unit” tables and their notes. Not public — but machine-readable in the dataset.

- OBP data availability is **inconsistent across parts** — ISO 80000-3 is there, ISO 80000-11 refers to other clauses
- Other unit datasets reference ISQ: **UnitsML and IEC CDD**

THE ASK

Who wants the dataset, and how

- **IEC SC 3D (CDD)**, via IEC TC 25 — reference data for electrotechnical quantities
- **ISO/TC 184/SC 4 (Industrial Data)** — quantity data inside industrial models
- **ISO/TC 154 (ebXML)** — semantic references in electronic business data
- Other TCs host their own terms — **ISO/TC 204** on GitHub, **ISO/TC 211** via its Maintenance Agency — we want the same autonomy for 80000 consumers

The answer is access: a stable, versioned, machine-readable service that any committee can build on — **without touching the standards themselves**.

L0
space

L1
length

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length of rod

L3
1.234 m ± 0.001

PART TWO

Entering SMART

The SDU SMART information model, and the ISO & IEC 80000 ontology built on top of it.

INFORMATION MODEL

The SMART information model is available

SDU SMART publishes a core ontology for standards data — generic classes for documents, terms, and their relations.

- Public at `smartsdu.org/informationmodel`
- ISQ Browser consumes it as a **downstream user** — the core stays untouched
- Domain extensions live in separate, versioned ontologies that import the core

`smart: core classes` `SHACL validation` `SKOS concepts`

ONTOLOGY

The ISO & IEC 80000 ontology, on top of “smart”

“isoiec80000” models what the series actually publishes — four domain classes over the SMART core:

Math notation
Signs and symbols of ISO 80000-2

Quantity
Quantities of Parts 3–13

Unit of measurement
Units with UnitsML identity

Dimension
Quantity dimensions from UnitsDB

- Sample instances published for each class — math, quantities, units, dimensions

L0
space

L1
length

L2
length of rod

L3
1.234 m ± 0.001

PART THREE

Modelling measurement itself

A four-level measurement hierarchy, implemented as linked data.

WHY AN EXTENSION

“length” alone cannot say what was measured

The published series defines *length* — the kind of quantity. It cannot express **the length of this rod, measured today, to ± 0.001 m**. That gap sits between committees and every industrial use of their data.

- **Pendrill (2024)** resolves this with a four-level hierarchy across three parallel axes: quantity, entity, measurement
- The ISQ extension implements the model as **OWL classes and properties** — not prose
- The published series stays untouched: the extension sits **beside** ISO 80000, pointing at it

Dybkaer (2010) · quantity axis

Fleischman (1960) · Sachgrösse

THE MODEL

Four levels, three axes

LEVEL	QUANTITY · DYBKAER	ENTITY	MEASUREMENT
0	kind of quantity *space*	—	—
1	quantity, Q = X *length*	kind of entity *physical body*	—
2	dedicated quantity *length of rod*	entity *rod*	—
3	quantity instantiation *length of rod AB, 09:15*	entity instantiation *rod AB*	quantity as measured *1.234 m ± 0.001*

Empty cells are part of the model — only the quantity axis reaches level 0; the measurement axis exists only at level 3. **Every cell is live in the browser's walkthrough.**

IMPLEMENTATION

The isq extension, v1.1.0

Classes — the hierarchy as OWL:

`isq:KindOfQuantity` `isq:Quantity` `isq:DedicatedQuantity`
`isq:QuantityInstantiation` `isq:MeasuredQuantity` `isq:EntityKind`
`isq:EntityConcept` `isq:PhysicalBody · Substance · Field · Wave · System`

Properties — the relations between levels:

`hasKind` `dedicatedTo` `characterizes` `instantiationOf` `instantiates`
`measurementOf` `instantiatedAt` `measuredAt` `hasValue` `uncertainty`

How it attaches to the series

- `isq:DedicatedQuantity` **dedicatedTo** → the published entry, e.g. `t3-1.1` “length”
- Entity kinds subclass SMART core entity classes
- Versioned `owl:versionInfo 1.1.0`, SHACL-validated in CI
- Served as RDF — Turtle and JSON-LD, downloadable per entity

WORKED EXAMPLE

One measurement, end to end: the rod

L1 · SERIES
t3-1.1 “length”
l
The published quantity — ISO 80000-3

→

L2 · DEDICATED
ex:lengthOfRod
length of rod
dedicatedTo t3-1.1 · characterizes ex:rod

→

L3 · INSTANTIATION
ex:lengthOfRodAB
rod AB, 09:15
instantiationOf ex:lengthOfRod · instantiates ex:rodAB

→

L3 · MEASURED
ex:measuredLengthOfRodAB
1.234 m ± 0.001
measurementOf · measuredAt · hasValue · uncertainty

Each step is a **live page** in the browser — and the same chain is downloadable as `isq-examples.ttl`.

GOVERNANCE

Where the extension lives — and what we never touch

Ours, versioned in the ISQ Browser

- `ontologies/isq.ttl` — the measurement layer, v1.1.0
- `ontologies/isq-examples.ttl` — the rod worked example
- Full ontology download at `…/ontology/full.ttl`

Theirs — consumed, never modified

- SDU SMART core ontology — **read-only upstream**
- ISO/IEC 80000 dataset — read-only source data
- UnitsDB (UnitsML) — read-only units and dimensions

All ISQ-specific modelling stays in **our namespace**. Upgrades to upstream sources flow in; nothing flows back that could destabilise them.

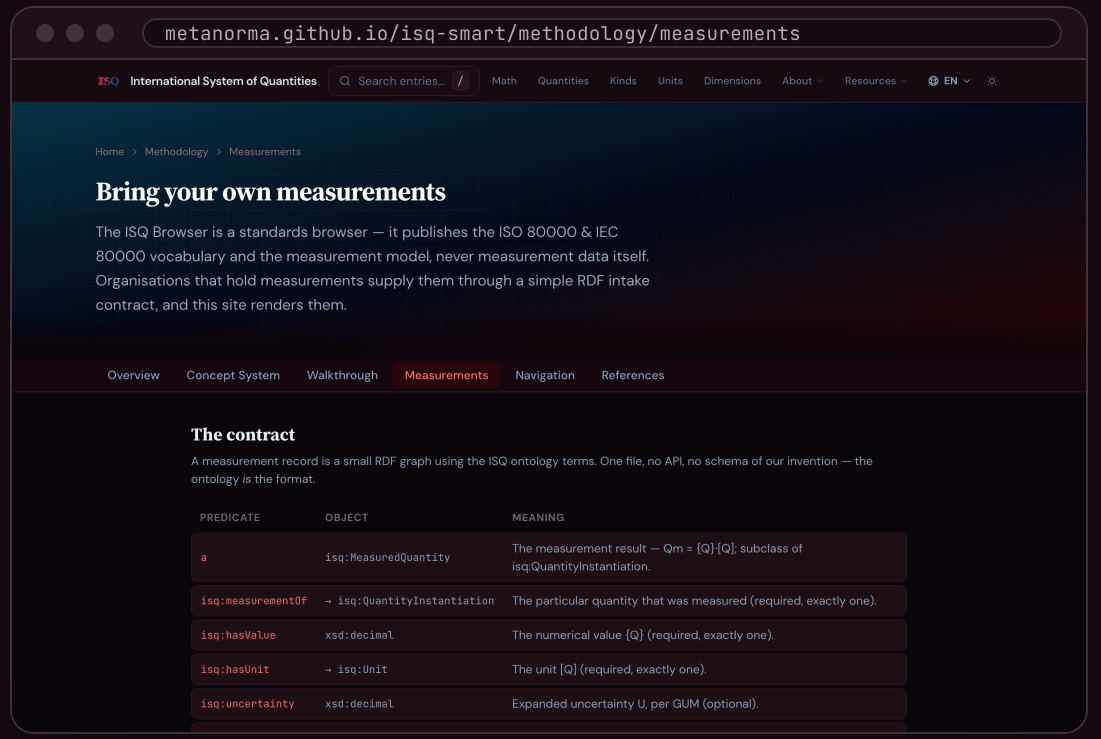
FOR DATA PRODUCERS

A contract for submitting measurements

The browser does **not** provide measurement data. It provides the **framework** — other bodies bring their own.

- Declare a dedicated quantity **dedicatedTo** a published entry
- Instantiate it against your entity, with time, value, uncertainty
- Validate with the published SHACL shapes; ship Turtle or JSON-LD

The intake path is documented at `…/methodology/measurements`.



▸ The measurement intake contract — framework, not data

L0–L3

the whole ladder, live

PART FOUR

The web service

The ISQ Browser — the series and its measurement layer, published as a static, linked-data site.

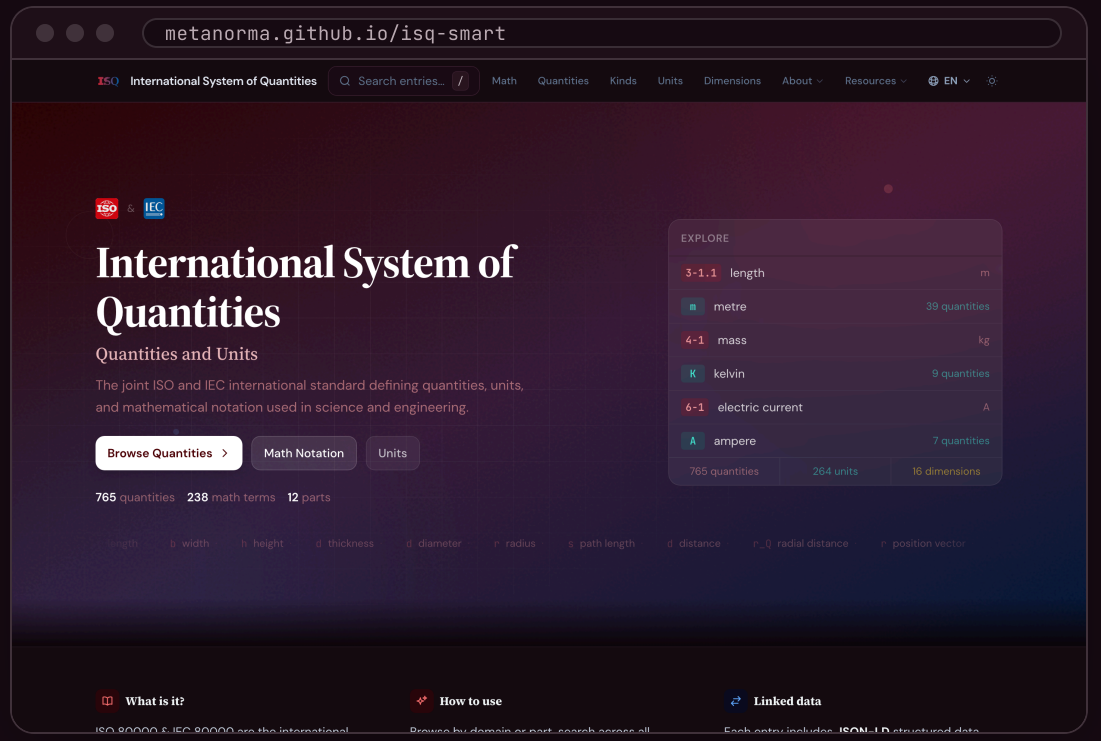
765 quantities | 230 math entries | 82 units | 18 dimensions | 242 ontology entities | 41 relations

WEB SERVICE

One front door to the series

- **Explore panel** — jump straight into quantities, math, units
- **Full-text search** across every entry — including ontology entities
- Dark and light themes, bilingual FR/EN, keyboard-first

metanorma.github.io/isq-smart

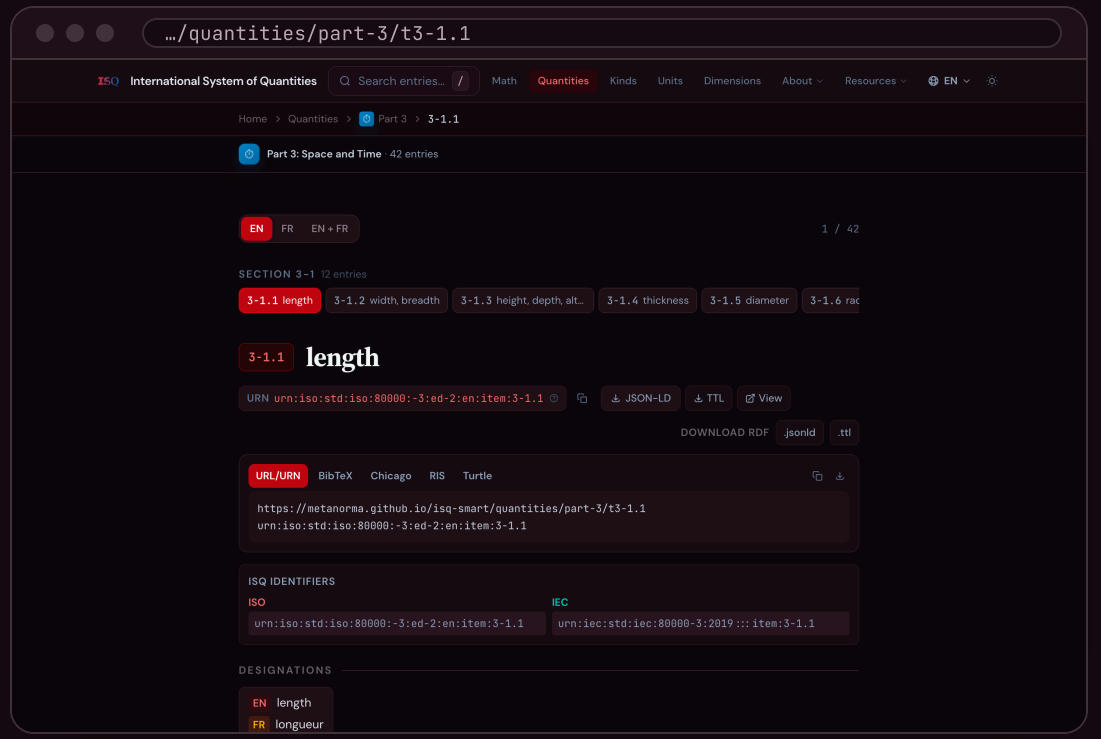


▸ The ISQ Browser home

QUANTITIES

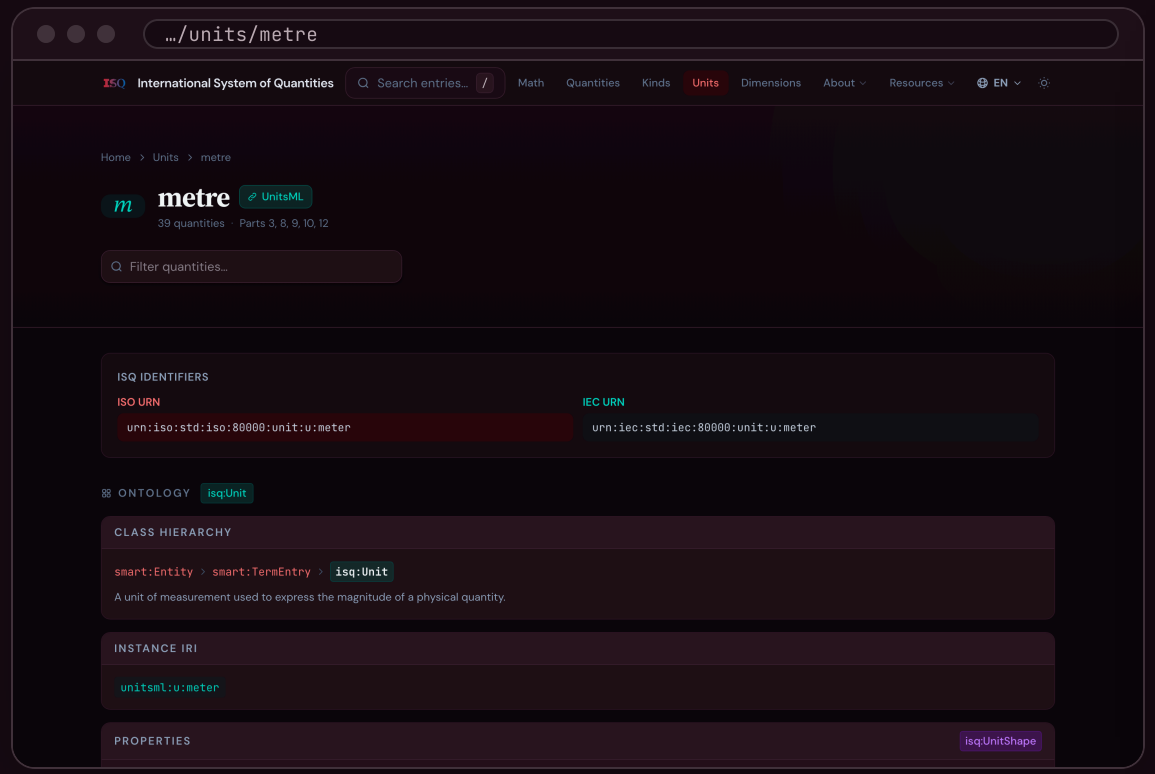
Every entry, fully linked

- **ID, name, symbol, definition, notes** — the normative tables, browsable
- Linked units, dimensions, math symbols and cross-references
- **Dedicated quantities** panel — the ontology layer attached to the published entry
- RDF downloads — Turtle and JSON-LD — per entry

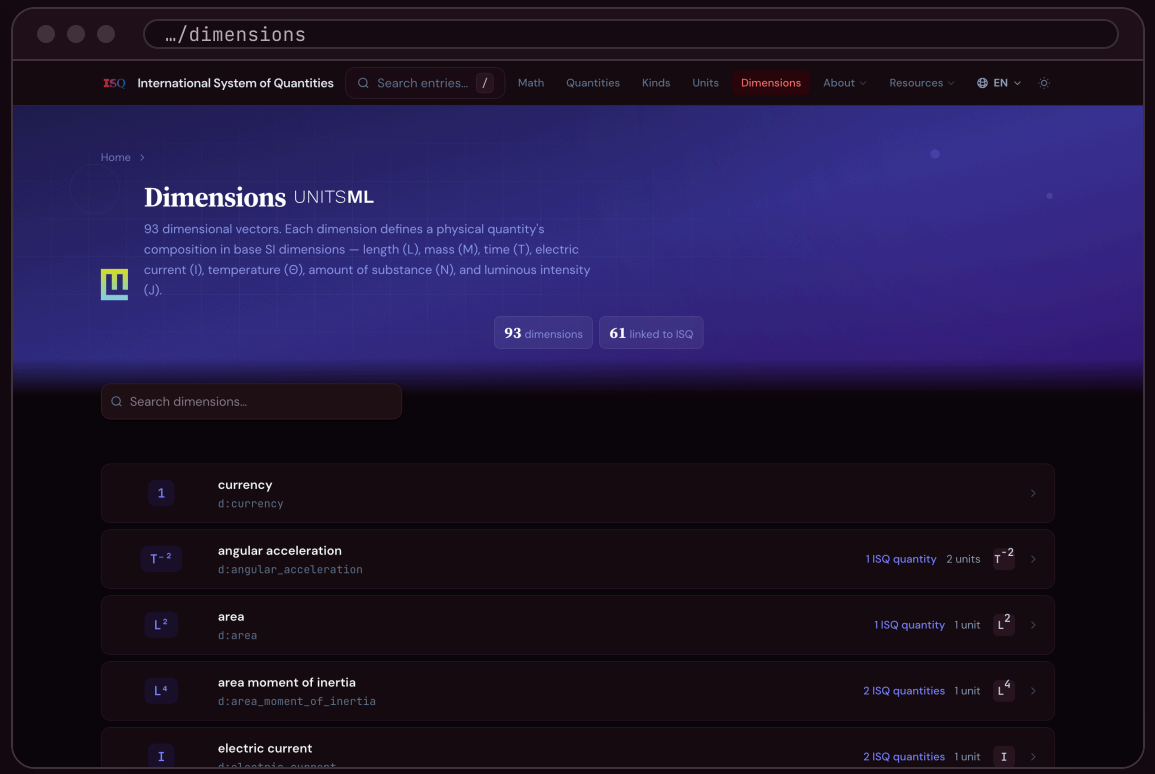


▸ t3-1.1 “length” — with its ontology attachments

Units and dimensions, from UnitsDB



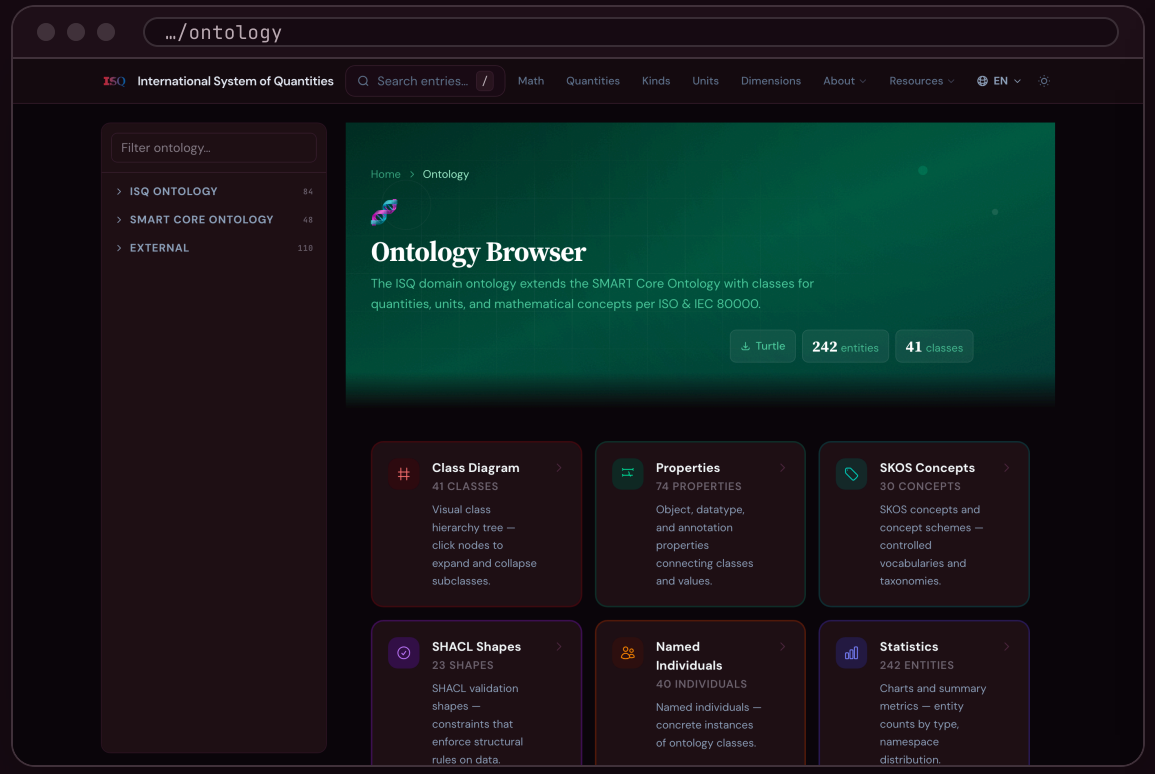
▸ 82 units with UnitsML identity and URNs



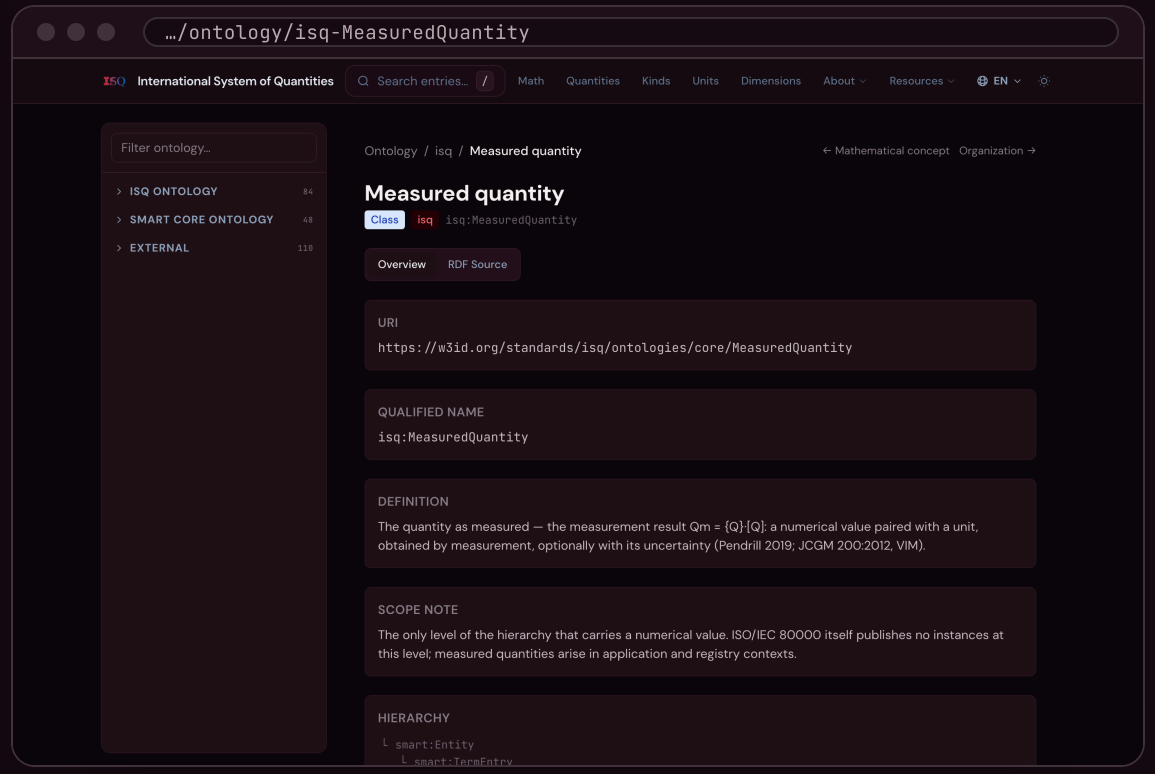
▸ 18 dimensions, sourced from UnitsML

ONTOLOGY BROWSER

The ontology, browsable



▸ 242 entities · 41 relations · SHACL shapes



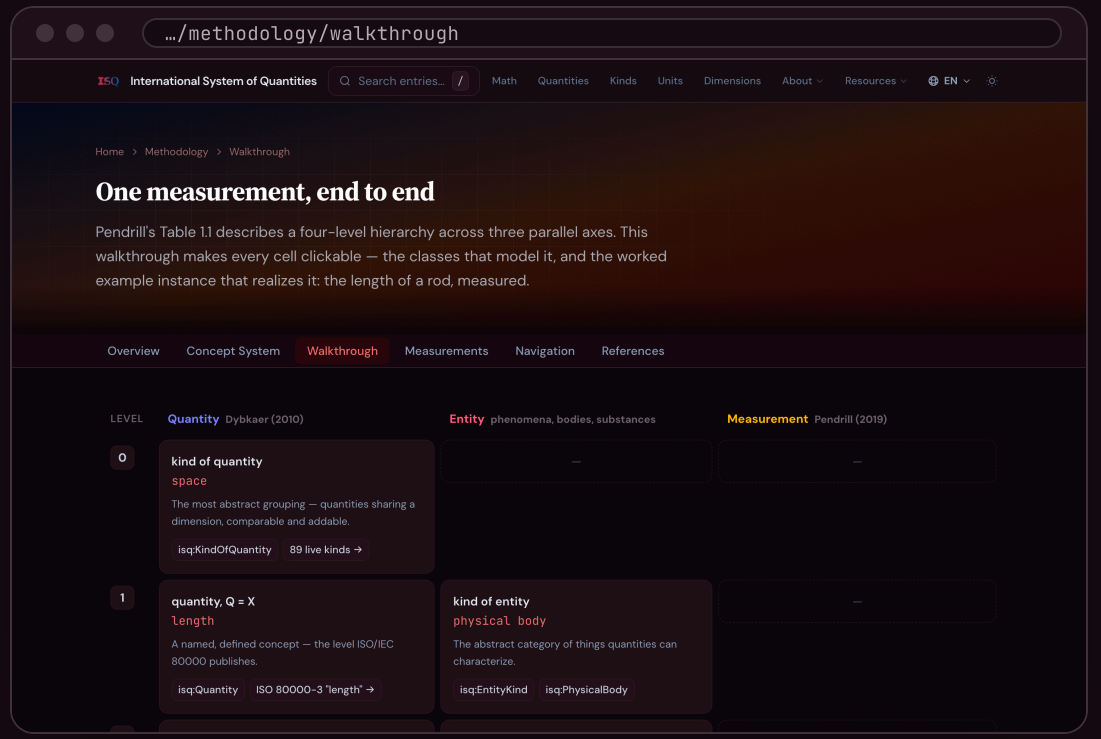
▸ Each class and individual gets a page — with RDF downloads

METHODOLOGY

The walkthrough — the hierarchy, live

- Every cell of the level table is **clickable** — classes and worked-example instances
- The four-step chain from “length” to **1.234 m ± 0.001**, as live pages
- Bilingual — the methodology reads in French too

…/methodology/walkthrough



▸ The measurement walkthrough — the ontology, demonstrated

PRODUCT DETAILS

Bilingual, searchable, downloadable

All French ISO and IEC parts encoded

- Every quantity name carries its French form
- One toggle — decimal commas rendered correctly
- Methodology pages fully translated

Search reaches the ontology

- Names, symbols, definitions, units — and `isq:MeasuredQuantity`
- Keyboard-first: `/` opens, arrows navigate, ↵ opens
- Lazy-loaded index — the site stays fast

And everything is data: **RDF exports per entity, per part, and for the whole ontology** — Turtle and JSON-LD.

ENGINEERING

Quality is part of the service

109k
links checked
every build, zero broken

0
type errors
blocking CI gate

608
automated tests
on every change

100%
static pages
no runtime server

- Ontology **SHACL-validated** at build time — broken data cannot ship
- Weekly external link audit; dependency updates automated
- Open source — `github.com/metanorma/isq-smart`

ISQ

Thank you.

Questions welcome — and the ladder is open for your data.

BROWSER	metanorma.github.io/isq-smart
DECK	…/isq-smart/deck/isq-tc12.html
WALKTHROUGH	…/methodology/walkthrough
SOURCE	github.com/metanorma/isq-smart

Ronald Tse — Ribose · A service contributed to ISO/TC 12 · November 2026

1.234 m ± 0.001